THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mass of 2 kg is raised by 3 m with an acceleration of 2 m/s squared from rest (g = 10). The work done by the lifting force is: 1

(a) 60 J **(b)** 36 J **(c)** 12 J **(d)** 72 J

2. The cost of running a 1.5 kW AC for 8 hours a day for 30 days at 7 rupees per unit is: 1

(a) 2520 rupees **(b)** 25200 rupees **(c)** 840 rupees **(d)** 84 rupees

3. Assertion (A): A fixed pulley has mechanical advantage equal to 1. Reason (R): It only changes the direction of the force. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Two bodies of masses m and 4m have equal KE. The ratio of their speeds is: 1

(a) 1 : 2 **(b)** 2 : 1 **(c)** 1 : 4 **(d)** 4 : 1

5. A motor of 10 kW lifts a mass of 500 kg at a constant speed. The speed is (g = 10): 1

(a) 2 m/s **(b)** 5 m/s **(c)** 20 m/s **(d)** 0.5 m/s

6. A pump of 2 hp (1 hp = 746 W) raises water at 80 per cent efficiency. The useful power is: 1

(a) 932 W **(b)** 1492 W **(c)** 746 W **(d)** 1194 W

7. Assertion (A): A machine can have an efficiency greater than 100 per cent. Reason (R): Output work cannot exceed input work. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A force of 10 N acts on a body that moves 4 m in the direction opposite to the force. The work done by the force is: 1

(a) 2.5 J **(b)** 40 J **(c)** 0 **(d)** -40 J

9. A car of mass 1000 kg moving at 72 km/h is brought to rest in 20 m. The average braking force is: 1

(a) 20000 N **(b)** 5000 N **(c)** 1000 N **(d)** 10000 N

10. The energy stored in a dam of 10^6 kg of water at 30 m (g = 10) is: 1

(a) 3 x 10^5 J **(b)** 3 x 10^8 J **(c)** 3 x 10^9 J **(d)** 3 x 10^7 J

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A man pulls a rope through 20 m with an effort of 50 N to raise a load of 100 N with an ideal machine. Find the distance moved by the load. 2

12. A 1000 W motor lifts 50 kg to 10 m. Find the time if the efficiency is 50 per cent (g = 10). 2

13. Why does a heavy truck need a long distance to stop? Show with KE = F x d for 10000 kg at 20 m/s and F = 50000 N. 2

14. Explain why power consumption of an electric appliance is written in watts and the energy in kWh on the bill. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The bob of a pendulum of mass 0.2 kg is released from a point 0.5 m above the lowest point. Find the maximum speed and the KE and PE when it is 0.2 m above the lowest point (g = 10). 3

16. A skier of mass 70 kg starts at rest at the top of a 50 m high slope and reaches the bottom with a speed of 25 m/s. Find the energy lost to friction and air (g = 10). 3

17. Explain, with the example of a ball on a hill, stable, unstable and neutral equilibrium in terms of potential energy. 3

OR

A body of mass 10 kg is pushed up a smooth 5 m long incline of height 2 m. Find the force needed, the work and MA (g = 10). 3

18. Explain why the work done in lifting a load through a given height by a lever, a pulley and an inclined plane is at least mgh. 3

OR

A pump raises 5000 kg of water to a tank 12 m high in 10 minutes. Find the power in watts and in hp (1 hp = 746 W) (g = 10). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how a roller-coaster demonstrates the conservation of mechanical energy, stating the minimum speed at the bottom of a 20 m drop (g = 10) and how friction changes the answer. 5

OR

A body of mass 4 kg is lifted by a rope from rest to a height of 10 m with an upward acceleration of 1 m/s squared. Find the tension, the work done by the tension, the work done by gravity, and the final KE (g = 10). 5

20. Compare an inclined plane (length 5 m, height 1 m), a lever (arms 1 m and 0.25 m) and a pulley system (VR = 4) in lifting a 400 N load: find the ideal effort in each, the work input and the common work output, and say which has the highest MA. 5

OR

Explain how energy is transformed in a coal power plant from the burning of fuel to electricity at home, with the typical efficiency of each step, and calculate the overall efficiency if boiler 80 per cent, turbine 40 per cent, generator 95 per cent. 5

END OF QUESTION PAPER — Total: 40 Marks